

# Vishal Kumar Allam

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## PROFESSIONAL SUMMARY

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- Results-driven **QA Automation / AI Operations Engineer** with **6+ years of experience** building software-quality, automation, evaluation, and production-validation solutions across healthcare applications, distributed systems, AI/ML platforms, and data-intensive enterprise environments.
- Hands-on experience designing and maintaining **automated evaluation pipelines for Generative AI and agentic AI systems**, covering single-turn, multi-turn, tool-use, retrieval-augmented generation (RAG), and end-to-end agent workflows.
- Strong programming expertise in **Python and pytest**, with experience developing reusable testing utilities, REST API automation, distributed-system validation, simulation environments, CI/CD quality gates, and production debugging tools.
- Experienced developing **golden datasets, agent benchmarks, LLM-as-judge evaluations, rubric-based scoring, regression detection, adversarial testing, prompt-injection probes, and human-review validation** for AI quality assurance.
- Hands-on experience with **LangGraph, LangChain, AutoGen, CrewAI, RAG pipelines, embeddings, vector search, tool calling, orchestration, guardrails, and human-in-the-loop controls**.
- Experienced using **LangSmith, Langfuse, Arize Phoenix, Ragas, DeepEval, and promptfoo** to measure task completion, tool-call accuracy, grounding, hallucination rates, response quality, latency, cost, safety, and production drift.
- Healthcare technology experience involving **PHI/PII, HIPAA, claims, payment integrity, OCI, Docker, Kubernetes, Terraform, SRE practices, observability, data governance, statistics, and production AI monitoring**.

## EDUCATION

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**University of Missouri Kansas City**

*"Master of Science" in Computer Science*

Kansas City, MO

*Jan. 2023 – May 2024*

**Jawaharlal Nehru Technological University**

*"Bachelor of Technology" in Computer Science*

Hyderabad, India

*Aug. 2015 - May 2019*

## TECHNICAL SKILLS

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- **Programming & Automation:** Python, pytest, Java, TypeScript, C#, SQL, REST APIs, JSON, Automation Tooling, Distributed Systems
- **Agentic AI & GenAI:** LLMs, Generative AI, Agentic AI, LangGraph, LangChain, AutoGen, CrewAI, Tool Calling, Orchestration, Guardrails, Human-in-the-Loop
- **LLM Evaluation:** Golden Datasets, Eval Datasets, LLM-as-Judge, Rubric-Based Scoring, Agent Benchmarking, Regression Detection, Task Completion, Tool-Call Accuracy, Grounding, Hallucination Metrics, Safety Evaluation
- **RAG & Observability:** RAG, Embeddings, Vector Search, Retrieval Evaluation, LangSmith, Langfuse, Arize Phoenix, Ragas, DeepEval, promptfoo, Log/Trace Analysis
- **AI Quality & Statistics:** Adversarial Testing, Prompt Injection, Negative Testing, Boundary Testing, Red Teaming, Drift Detection, Sampling, Variance, Statistical Significance, Latency and Cost Analysis
- **Cloud & DevOps:** Oracle Cloud Infrastructure (OCI), AWS, Azure, Docker, Kubernetes, Terraform, Infrastructure as Code, CI/CD Quality Gates, Production Monitoring, SRE Practices
- **Healthcare & Governance:** HIPAA, PHI, PII, Claims, Payment Integrity, Data Governance, Auditability, AI Safety, Quality Dashboards, Runbooks, RCA, SWE-bench, GAIA, AgentBench, tau-bench

## EXPERIENCE

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### Test Automation Engineer *PCMI LLC (Full Time)*

Oct 2024 – Present  
*Chicago, IL, USA*

- Designed and maintained automated **agentic AI test suites and evaluation pipelines** using Python and pytest to validate single-turn, multi-turn, tool-use, RAG, and end-to-end workflows across production-oriented AI-enabled applications.
- Developed and curated **golden datasets, benchmark scenarios, simulation environments, synthetic inputs, and edge-case suites** that reflected real business workflows and enabled repeatable comparison across model, prompt, retrieval, and tool changes.
- Implemented **LLM-as-judge and rubric-based scoring** to measure task completion, tool-call accuracy, grounding, hallucination rate, response quality, latency, cost per task, and safety compliance, correlating automated scores with human-review outcomes.
- Integrated agent evaluations into **CI/CD pipelines** with automated regression detection and explicit pass/fail quality gates, preventing degraded model, prompt, retrieval, or tool configurations from progressing to production.
- Triaged and root-caused complex agent failures by analyzing **LLM traces, prompts, tool invocations, retrieval steps, orchestration paths, API calls, logs, and distributed-system behavior** using LangSmith, Langfuse, and Arize Phoenix.
- Performed adversarial and negative testing using prompt-injection probes, malformed inputs, boundary conditions, unsafe tool scenarios, retrieval failures, hallucination tests, and guardrail-verification suites to strengthen agent safety and reliability.
- Built RAG and agent-evaluation workflows using **LangGraph, LangChain, AutoGen, CrewAI, embeddings, vector search, Ragas, DeepEval, and promptfoo**, assessing retrieval relevance and its downstream impact on agent responses.
- Partnered with AI Engineers, Data Scientists, Product, and DevOps teams to improve agent observability, testability, PHI/PII controls, HIPAA compliance, production monitoring, drift detection, dashboards, runbooks, and human-in-the-loop workflows.

**Environment:** Python, pytest, Generative AI, Agentic AI, LLMs, LangGraph, LangChain, AutoGen, CrewAI, RAG, Embeddings, Vector Search, Golden Datasets, LLM-as-Judge, LangSmith, Langfuse, Arize Phoenix, Ragas, DeepEval, promptfoo, REST APIs, OCI, AWS, Docker, Kubernetes, Terraform, CI/CD, PHI, PII, HIPAA, RCA, Quality Dashboards

### SDET

Dec 2023 – Sept 2024  
*Remote, USA*

*Client: CVS*

- Built automated evaluation and regression suites for **AI/ML and Generative AI-enabled healthcare applications**, validating conversational agents, RAG workflows, clinical and claims-oriented decision support, workflow automation, and tool-invocation scenarios.
- Developed healthcare-focused **golden datasets and simulation scenarios** covering claims, payment integrity, payer workflows, PHI/PII handling, edge cases, unsafe actions, retrieval failures, and human-in-the-loop review requirements.
- Automated agent benchmarking using Python and pytest, measuring **task completion, grounding, hallucination rate, response quality, tool-call accuracy, latency, cost per task, and safety compliance** across model and prompt revisions.
- Implemented **LLM-as-judge, rubric-based scoring, Ragas, DeepEval, and promptfoo** evaluations and validated automated scoring against human and domain-review outcomes to improve evaluation consistency and reliability.
- Performed root-cause analysis of agent failures using LangSmith/Langfuse traces, vector retrieval results, LLM prompts, tool execution logs, REST API calls, orchestration paths, and application telemetry, producing actionable defect reports.
- Validated **RAG pipelines, embeddings, and vector search** by measuring retrieval relevance, grounding, context precision/recall, citation consistency, and downstream response quality across sensitive healthcare workflows.
- Executed adversarial and red-team scenarios covering prompt injection, malformed input, boundary conditions, unsafe actions, guardrail bypass attempts, sensitive-data leakage, and unexpected tool behavior.
- Integrated agent tests into CI/CD and production monitoring workflows while validating **HIPAA, PHI/PII controls, auditability, data governance, Kubernetes/Docker deployments, production drift, and quality dashboards**.

**Environment:** Python, pytest, Agentic AI, Generative AI, LLMs, RAG, LangGraph, LangChain, Golden Datasets, LLM-as-Judge, LangSmith, Langfuse, Ragas, DeepEval, promptfoo, Embeddings, Vector Search, REST APIs, OCI, AWS, Docker, Kubernetes, CI/CD, Claims, Payment Integrity, PHI, PII, HIPAA

## Software Engineer

Sept 2021 – Dec 2022

*Infor*

*Hyderabad, India*

- Developed Python and Java-based automation frameworks and backend validation utilities for enterprise applications, distributed services, APIs, and data-intensive workflows.
- Built reusable **pytest-based automated suites** for API, integration, regression, negative, data-validation, and end-to-end scenarios, improving early detection of production-impacting defects.
- Created structured test datasets, reusable fixtures, simulation inputs, boundary-value scenarios, malformed-input tests, and regression baselines to validate complex application and service behavior.
- Performed root-cause analysis using application logs, API traces, SQL queries, service responses, and execution history to reproduce intermittent, distributed, and environment-specific failures.
- Integrated automated testing into Git, Jenkins, Maven, Docker, and CI/CD workflows, implementing quality gates and repeatable regression validation before release promotion.
- Developed quality dashboards and reports covering pass rates, execution stability, defect trends, regression results, release readiness, and statistical comparisons across test runs.
- Validated data quality, schema consistency, transformation logic, API contracts, transaction processing, and backend integrity across relational databases and distributed enterprise services.
- Collaborated with Software Engineers, Product Managers, QA teams, and DevOps to improve testability, observability, automation coverage, defect documentation, debugging practices, and production readiness.

**Environment:** Python, pytest, Java, REST APIs, Distributed Systems, SQL, Data Validation, Regression Testing, Negative Testing, Jenkins, Git, Maven, Docker, CI/CD, Log Analysis, Trace Analysis, RCA, Quality Metrics, Statistical Analysis, Agile Scrum

## Associate Engineer QA

June 2019 – August 2021

*Virtusa*

*Hyderabad, India*

- Developed reusable automation frameworks for enterprise web, mobile, API, and backend applications using Java, Python, Selenium WebDriver, Appium, Cucumber BDD, and TestNG.
- Designed and executed functional, integration, regression, negative, boundary, system, and end-to-end test scenarios to identify software and integration failures before production deployment.
- Built API-validation suites covering request/response contracts, malformed inputs, error handling, authentication, business rules, service dependencies, and downstream data consistency.
- Created reusable test datasets and environment configurations to reproduce complex workflows, edge cases, integration failures, and regression scenarios across multiple application releases.
- Investigated defects through application logs, API responses, SQL queries, service behavior, and execution history, supporting detailed root-cause analysis and reproducible defect reporting.
- Integrated automation suites with Jenkins, Git, Maven, and CI/CD pipelines to provide continuous regression testing, repeatable validation, automated quality gates, and release-quality feedback.
- Produced quality reports and dashboards summarizing test execution, pass/fail trends, defect severity, regression risk, coverage gaps, and release-readiness recommendations.
- Collaborated with Software Engineers, Business Analysts, QA leads, and Product teams to improve system testability, strengthen automation coverage, resolve complex defects, and document operational testing procedures.

**Environment:** Java, Python, Selenium WebDriver, Appium, Cucumber BDD, TestNG, REST APIs, SQL, Negative Testing, Boundary Testing, Regression Testing, System Testing, End-to-End Testing, Jenkins, Git, Maven, CI/CD, Log Analysis, RCA, Quality Reporting

## AWARDS & ACHIEVEMENTS

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- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.